



Data Structures and Algorithms -3 (25CS2103E)

Threat-Intelligence Kill-Chain Correlation Engine

Team Members

SRIVALLI:2520090040

PAVANI:25200090041

KAVYASREE:2520090143

Under the Esteemed Guidance of

MS.Dr. DYUTI BANERJEE

**ASSISTANT
PROFESSOR, AI&DS**



Abstract

The Threat-Intelligence Kill-Chain Correlation Engine is a cybersecurity analytics system that combines threat intelligence with the Cyber Kill Chain framework to detect and analyze cyber attacks. The project uses data structures and algorithms to efficiently process large volumes of security events and identify relationships between attack indicators. Security logs, alerts, and threat intelligence feeds are collected and correlated to determine the progression of an attack across different kill-chain stages. Graph-based analysis, hashing techniques, and priority-based processing are used to improve correlation accuracy and reduce response time. The system provides real-time visualization of attack paths, highlights critical threats, and assists security teams in making informed decisions. By automating threat correlation and attack tracking, the proposed solution enhances situational awareness, reduces false positives, and enables proactive defense against advanced cyber threats. This project demonstrates the practical application of DSA concepts in solving real-world cybersecurity challenges.

Introduction

With the rapid growth of cyber threats, organizations face significant challenges in identifying, analyzing, and responding to security incidents in real time. Traditional security systems often generate large volumes of alerts, making it difficult for security teams to determine the actual attack path and prioritize responses. The Threat-Intelligence Kill-Chain Correlation Engine is a Data Structures and Algorithms (DSA) based project designed to correlate security events with cyber threat intelligence using the Cyber Kill Chain model. The system collects and organizes threat data from multiple sources, such as intrusion detection systems, firewalls, and threat intelligence feeds. Using efficient data structures like graphs, hash tables, and priority queues, the engine maps attack indicators to different stages of the kill chain, including reconnaissance, weaponization, delivery, exploitation, installation, command and control, and actions on objectives. Advanced algorithms analyze relationships between events, detect attack patterns, and provide a clear view of the attack lifecycle. This helps security analysts identify threats faster, improve incident response, and strengthen overall network security.

Literature Survey

Several studies have focused on threat intelligence and attack correlation techniques to improve cybersecurity defenses. Traditional intrusion detection systems can detect suspicious activities but often generate a large number of alerts without providing attack context. Researchers have proposed the Cyber Kill Chain framework to understand attacker behavior through different stages of an attack. Recent advancements in threat intelligence platforms have enabled organizations to collect and analyze threat data from multiple sources. Graph-based correlation and machine learning techniques have further improved the ability to identify complex attack patterns. However, many existing systems lack efficient event correlation and real-time attack tracking, creating a need for an intelligent kill-chain correlation engine.

Problem Statement and Objectives

Problem Statement

Organizations face difficulties in identifying the complete lifecycle of cyber attacks due to fragmented security alerts and large volumes of threat data. Existing systems often fail to correlate events effectively, resulting in delayed detection and response.

Objectives

To collect and analyze security events from multiple sources.

To map attack indicators to Cyber Kill Chain stages.

To correlate related events using efficient DSA techniques.

To visualize attack progression and threat relationships.

To improve threat detection accuracy and incident response time.



Hardware and Software requirements

Hardware Requirements

Processor: Intel Core i3/i5 or above

RAM: 4 GB minimum (8 GB recommended)

Storage: 100 GB free disk space

Network Connectivity

Software Requirements

Operating System: Windows/Linux

Programming Language: Python

Database: MongoDB / MySQL

Libraries: NetworkX, Pandas, NumPy

IDE: VS Code / PyCharm

Visualization Tools: Matplotlib, Plotly



Conclusion

The Threat-Intelligence Kill-Chain Correlation Engine provides an effective solution for analyzing and correlating cyber threat data. By combining threat intelligence with the Cyber Kill Chain framework, the system helps security analysts understand attack behavior and respond quickly to threats. The use of data structures and algorithms improves processing efficiency and enables real-time threat analysis. The project contributes to stronger cybersecurity monitoring and incident management.

Future Scope

- Integration with AI and Machine Learning for predictive threat detection.
- Real-time threat intelligence feed integration.
- Cloud-based deployment for large-scale environments.
- Automated incident response and mitigation mechanisms.
- Advanced visualization dashboards for security analysts.
- Support for emerging cyber threat frameworks and standards.

Thank you